



EXECUTIVE IPSEC SECURITY BRIEFING

Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

SECURITY HARDENED PASS

NONE · COMPLIANT

Link 'Link 1: HQ Gateway <-> Datacenter Core' demonstrates complete cryptographic resilience under NIST SP 800-77 Rev. 1 and NSA CNSA 2.0 specifications. Strict AEAD authentication (AES-256-GCM), authenticated DH key exchange (Group 19 (ECP-256)), and monotonic sequence counters verified with zero defects.

SESSION TIMESTAMPS

15.0 sec

Live Link Telemetry: +0.0s to +15.0s (15.0 sec)

ORIGIN GATEWAY

**Link 1: HQ Gateway <-> Datacenter Core
(Initiator Node)**

172.28.0.2 (HQ Gateway) · StrongSwan Client Daemon

SECURED TUNNEL

IPsec TUNNEL Tunnel · AES-256-GCM

AES-256-GCM / Group 19 (ECP-256)

REMOTE GATEWAY

**Enterprise Responder Node (Link 1: HQ
Gateway <-> Datacenter Core)**

172.28.0.3 (DC Core) · StrongSwan Server Gateway

Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

COMPOSITE RISK INDEX



NIST SP 800-77 REV. 1 COMPLIANCE

PASS

NIST SP 800-77 Rev. 1

CRYPTOGRAPHIC RESILIENCE

100%

AES-256-GCM + DH
Group 19 Active

COMPLIANCE ALIGNMENT

100%

NIST SP 800-77 &
CNSA 2.0 Compliant

DEPLOYMENT READINESS RATING

READY FOR PRODUCTION

Fully compliant with federal defense-in-depth criteria and
ready for mission-critical deployment.

THREAT EXPOSURE INDEX

LOW

Zero active security
gaps

CHANNEL STABILITY

99.9%

Monotonic sequence
verification



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

Wiretap Eavesdropping & Decryption Sev: LOWLikelihood: LOW Zero risk. Combined authenticated encryption immunizes against eavesdropping and padding oracle attacks. PROTECTED Key Compromise & Retrospective Decryption Sev: LOWLikelihood: LOW Zero risk. Ephemeral secondary DH exchange on Child SAs guarantees complete Perfect Forward Secrecy. PROTECTED
Credential Cracking & Impersonation Sev: LOWLikelihood: LOW Zero risk. Mutual authentication with SHA-256 PRF prevents identity spoofing and dictionary cracking. PROTECTED Replay Injection & Sequence Tampering Sev: LOWLikelihood: LOW Zero risk. RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted across connection lifetime. PROTECTED



TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



Executive Sign-Off & Platform Attestation



